Autonomous Image Cartoonification Using Convolution and Toeplitz Matrix-Based Filtering in Robot Vision

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***Abstract*—Cartoonification of Images is a process of converting natural images into visualisation of images which are simple in nature but contain informative structural details without revealing the textual information present in the images. In the paper, the concept of an autonomous system of real-time cartoonification of images is introduced. The entire process is mathematically based and is used in the background of a robot vision. The entire process is based on a series of processes such as the detection of an object using the YOLOv8 algorithm, the weighted prioritization of the region of interest, the Toeplitz matrix convolution filtering, the gradient edge detection, and the K-Mean color quantization. The convolution is replaced by the LSI transformation which is represented in the form of a Toeplitz matrix. A composite scoring function based upon normalized bounding box area, spatial centrality to the center of the frame, and detection confidence is employed in order to determine the areas in a manner that minimizes redundant computation for approximating robotic visual attention. Normalized convolution kernels are employed in order to smooth out the high-frequency content of the image. Gradient operators are employed in order to maintain structural edges. Clustering-based optimization is employed in order to simplify the color space and obtain a flat shading effect characteristic of cartoons. Quantitative metrics are employed in order to determine the performance of the system in terms of variance reduction, color space simplification ratio, and mean squared error when compared to the original image. The quantitative metrics and experimental results attest towards the fact that a good balance of abstraction in the framework is achieved.**

***Index Terms*—Image Processing, Convolution, Toeplitz Matrix, Cartoonification, Robot Vision, YOLOv8, K-Means Clustering, Shift-Invariant Systems**

I. Introduction

The problem of transforming photos to cartoons is essentially a problem of signal processing, linear algebra, and computer vision. A digital image is essentially a matrix of pixel values; it is a two-variable function that is well suited for operations like convolution, vectorization, and spectral analysis [1], [12], [13]. Algebraic methods of image processing are used here, which eliminates any room for guessing.

The problem with transforming an image to a cartoon is that it is essentially a balancing act that tries to remove sufficient detail to make it look drawn while leaving sufficient detail to make it recognizable. It is necessary to remove texture, smooth out color transitions, but preserve contours that define shapes. Earlier attempts at non-photorealistic rendering of images were essentially attempts to combine smoothing filters with edge enhancement. Such images were surprisingly realistic given the simple processing [21]–[23]. Bilateral filtering and its variants essentially proved that it is possible to smooth out detail in flat areas while leaving edges essentially intact [3], [24], [25]. This is essentially how later research has proceeded.

The drive to create noise-resistant processing has led to the development of adaptive Gaussian methods that preserve structural detail even in the face of real-world noise [31]. Deep convolutional networks have revolutionized what is possible in image understanding systems [6], [7], [15]. Style transfer methods extended the concept of reinterpretation by training systems to recreate photographs in a particular style [4], [28], [29]. The resulting images may be breathtakingly beautiful, but the computational overheads made these systems impossible to use in practice.

Another trend, perhaps not as well-known in the world of image stylization, may be the increasing use of physics-informed and mathematically structured neural network architectures to solve problems in stochastic dynamics and integral

equations [36], [37]. This body of work suggests that there may be a trend developing that mathematical rigor and data flexibility are not at odds, and that designing a learning system that is both tractable and interpretable may be the key to better understanding and generalization.

Our paper is an extension of this idea. Rather than applying cartoonification as a problem of pattern matching to be learned, we view it as a process that consists of a set of linear operations that classical system theory is able to analyze completely. Convolution, which is part of the smoothing process, is reformulated as a linear shift-invariant operator and is expressed in Toeplitz matrix form. This allows us to analyze its effects analytically. We use YOLOv8 [2], [19] to incorporate knowledge of objects. It is used to direct our weighted scoring function toward those parts of the image that are most salient. It is efficient and interpretable, and it is able to operate in real-time, which makes it especially suited for use with robots.

II. Related Work

Image abstraction and cartoon rendering. However, the problem of rendering photographs to stylized images has been a focus of much research in recent decades. Winnemöller *et al.* [21] showed that parallel execution of an edge-preserving smoother with a difference-of-Gaussians edge detector was enough to create real-time video abstraction that appeared to be hand-drawn. Wang and Hu [22] surveyed this area a few years after that, identifying two broad categories of solutions: those based on filtering and those based on segmentation, although neither approach is entirely satisfying on its own. In particular, in the filtering approach, Kyprianidis *et al.* [23] showed that anisotropic Kuwahara filtering could suppress high frequency texture while leaving structural edges visually crisp, a property that also applies to the bilateral filter [3] and its variants such as domain transform [24] and edge-preserving decomposition [25]. More recent work by Aarthi *et al.* [31] extended this approach with a new class of adaptive Gaussian filters that were robust to sensor noise, which is a problem for video captured using low-cost cameras on robot arms.

Object detection and visual attention. However, accurate detection of objects in a scene is a prerequisite for any form of region-based image stylization. While handcrafted feature descriptors such as SIFT [8] and SURF [9] were early benchmarks for keypoint-based image recognition, deformable part models [10], [17] brought spatial structure to image detection pipelines. Deep learning has dramatically changed our notion of detection accuracy; AlexNet [6], ResNet [7], VGGNet [15] have been a series of accuracy records for image classification on ImageNet, while COCO [18] has provided a unified benchmark for measuring image detection accuracy. However, for image detection in robotics, accuracy is not enough; it also needs to be efficient enough to run in real-time. Single-shot image detection networks in YOLO [2], [19] have provided a new set of tools for real-time image detection on resource-constrained systems.

Learning-based perception in engineering domains. Outside of vision, these models have also been shown to be effective in other diagnostic and monitoring scenarios. For example, random forest regression models [32], GRU-based models [33], and CNN-BiLSTM models [34] have all been used for structured signal prediction problems. With regard to medical image processing, Karthikeyan *et al.* [35] demonstrated that models based on convolutional architectures could be used to detect infected cells in microscopy images of malaria. From an even more abstract standpoint, Singh and Sarkar [36] and Singh [37] have used neural solvers as part of physics-informed models for stochastic and integral equations, showing that mathematical interpretability and neural networks are not necessarily mutually exclusive

Gap addressed by this work. What the above body of literature seems to be lacking is a framework in which detection and stylization are not treated in a serial fashion in a somewhat ad hoc way, but rather in a unified framework in which the region of the image in which the stylization should be applied is a mathematically well-defined choice. The state-of-the-art in cartoon rendering software is either the application of a uniform effect across the entire frame or manual specification of the region of interest. The framework developed in the present work addresses this need through the application of a weighted scoring function on the YOLOv8 object detections to select the most visually salient object, and the application of the Toeplitz matrix smoothing, edge detection, and K-Means quantization techniques only on the region of the image in which the salient object appears.

III. Mathematical Background

In this section, the mathematical foundations of the proposed cartoonification system will be developed. The framework is based on discrete signal processing and linear systems, as well as clustering-based optimization. This way, each stage of the processing can be mathematically defined.

A. Image as a Matrix

A grayscale image of size $m \times n$ is represented as a discrete function in two variables, where each pixel p_{ij} represents image intensity at position (i, j):

$$I = \begin{bmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mn} \end{bmatrix} \tag{1}$$

Such a matrix structure allows us to take advantage of conventional linear algebraic operations. A colour image is represented as a sum of three separate channel matrices to ensure colour consistency:

$$I = \{I_R,\ I_G,\ I_B\} \tag{2}$$

B. Convolution

The discrete convolution operation is the basic filtering operation. Given an image I and a filter K, the output Y of the filter at position (i, j) is given by :

$$Y(i,j) = \sum_m \sum_n I(i-m,\, j-n)\, K(m,n) \tag{3}$$

where I refers to the image under consideration, K refers to the convolution kernel, and Y refers to the filtered image. Gaussian-type kernels are used for noise reduction of high frequencies while retaining major structural edges.

$$\mathcal{H}(aI_1 + bI_2) = a\mathcal{H}(I_1) + b\mathcal{H}(I_2)$$

The linearity property proves that convolution indeed forms a linear transformation and can be represented in matrix form.

C. Toeplitz Representation

If the input image I is vectorized to a column vector $\mathbf{x}$, then the convolution operation may be written equivalently as:

$$\mathbf{Y} = T\mathbf{x} \tag{4}$$

where T is a Toeplitz matrix whose elements take values from the kernel coefficients. A Toeplitz matrix has the property that its elements along any diagonal have the same value. This reflects the fact that the kernel coefficients are applied uniformly over the entire image domain, which is the defining characteristic of a linear shift-invariant system. This form of the convolution operation shows that filtering using convolution is not simply a localized averaging process, but a global linear operation on the entire image vector.

D. K-Means Objective

The characteristics of the cartoon images are the flat homogeneous color regions. To obtain this property, the color quantization is performed by minimizing the total variance within the clusters. Mathematically, the equation is represented as follows:

$$\min \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \tag{5}$$

where C_i is the ith cluster and $\boldsymbol{\mu}_i$ is the centroid of the ith cluster. The number of colors is reduced to k, which compress

IV. Proposed System Architecture and Methodology

Four loosely-coupled modules are used to perform the end-to-end process from the raw video frame to the final cartoon output, namely, an image acquisition front-end, an object detection and selection module, a multiple-step stylization process, and an evaluation back-end. This loose-coupling has the advantage that the individual modules can be profiled, tuned, and even changed independently of the others, and it is also easy to plug in GPU-based implementations of the more computationally intensive numerical steps in the near future.

A. System Workflow

The process is strictly left-to-right, comprising the following seven steps:

1) **Image Acquisition:** The live frame is read from the camera, stored as an RGB image of size $H \times W \times 3$.
2) **Object Detection:** YOLOv8 makes a single forward pass through the image, returning bounding box, class, and confidence predictions.
3) **Object Selection:** Each box is scored based on spatial size, distance from center, and confidence. The box with the highest score is chosen for stylization.
4) **Convolution Smoothing:** A Toeplitz-encoded Gaussian filter smooths the chosen region, preserving large-scale features but removing high-frequency texture.
5) **Edge Detection:** A Canny edge operator is used to detect binary edges, which will be used to generate the bold edges seen in cartoons.
6) **Colour Quantization:** K-Means clustering is used to reduce the number of colors in the image to k uniform patches, effectively "flattening" gradients.
7) **Edge Masking:** The edge map is overlaid onto the quantized image to produce the stylized image.

Each of these operations has a clear mathematical form, making the entire process transparent, easy to analyze, and straightforward to implement.

B. Image Acquisition

The video frame received is a matrix of size $H \times W \times 3$, where each matrix is divided into red, green, and blue color components. Before it is passed through the detector, it can be resized to match the input resolution expected by YOLOv8 and normalized to lie within the range $[0, 1]$—two simple preprocessing steps that help stabilize performance across cameras with different sensor characteristics. The fact that we are working with one frame at a time makes it easy for it to fit into a streaming loop.

C. Object Detection and Selection

YOLOv8 makes a single forward pass through the preprocessed image, simultaneously regressing the coordinates of bounding boxes and predicting class probabilities for all regions of interest. The output for an image frame may be represented as $(\{(b_i, c_i, s_i)\}_{i=1}^{N}$, where $b_i = (x_{1i}, y_{1i}, x_{2i}, y_{2i})$ are the coordinates of the corners of the ith box, c_i is the predicted class for the ith box, and s_i is the confidence score for the ith box, where s_i is between 0 and 1. Non-Maximum Suppression is then used to merge heavily overlapping bounding boxes, keeping only the box with the highest score in each group of detections. For environments where multiple objects are present, as is often the case in uncontrolled environments, stylizing all detections simultaneously would be counterproductive. The scoring function, as discussed in the following subsection, will be used to select one of the regions of interest.

D. Object Selection Scoring

The scoring mechanism integrates geometry and probability to form a single deterministic composite score. Since there may be multiple objects that appear at once in one frame, the system needs to identify the most important target object to be stylized. Instead of using arbitrary selection, a mathematically defined scoring function is proposed to enable deterministic decision-making. The proposed scoring mechanism mimics the concept of smart vision in robotic perception by using spatial salience, centrality, and reliability in one score.

Area Component:

$$A_i = (x_{2i} - x_{1i})(y_{2i} - y_{1i}) \tag{6}$$

The area component represents the spatial size of each detected bounding box. Generally, objects with larger areas are closer to the camera or more salient. The raw area is normalized with respect to the total size of the frame. This ensures that all detected objects are treated equally regardless of image resolution.

Centrality Component:

$$C_i = 1 - \frac{d_i}{d_{\max}} \tag{7}$$

where d_i represents the Euclidean distance between the center of each bounding box and the center of the frame, and $d_{\max}$ represents the maximum possible value for such a distance. The centrality scores are higher for objects that are closer to the optical axis.

Confidence Component: The object detection confidence score, denoted as s_i, represents the confidence level with respect to object detection using the YOLOv8 model. Higher confidence scores represent higher levels of certainty that the detected bounding box corresponds to a valid object class. The confidence score ensures that weak object detections are not given undue weight, even if they are large and central.

Final Weighted Score:

$$\text{Score}_i = 0.4\, A_i^{\text{norm}} + 0.3\, C_i + 0.3\, s_i \tag{8}$$

The weights balance the object's geometric significance (area), spatial topicality (centrality), and detection reliability (confidence). The object with the highest value is chosen for stylization:

$$i^* = \arg\max_i \ \text{Score}_i \tag{9}$$

This deterministic selection rule ensures that object selection is repeatable and uniform. The object selection module combines geometric, spatial, and probabilistic information, thereby simulating visual attention in robotic vision while avoiding unnecessary computation on less significant objects.

E. Stylization Pipeline

Once the highest-scoring object is identified, a four-stage stylization process converts the selected region into a stylized cartoon-like image that minimizes texture and maximizes structural details.

Smoothing with Convolution: The Gaussian-based Toeplitz convolution filter is a linear shift-invariant low-pass filter that reduces texture and noise while maintaining the dominant structural features. The image brightness is normalized through a mean normalization process. This process can be viewed as a mathematically well-founded low-pass filtering process based on LSI system theory.

Edge Detection: The structural boundaries are strongly defined by employing a gradient operator that is sensitive to changes in pixel intensity. Thresholding is then performed on the gradient map, which allows for the extraction of prominent contours, thereby defining the bold line appearance of cartoons.

K-Means Colour Quantization: A down-sampling of the color space of the selected region is performed to obtain k clusters. Pixels are then assigned to their closest cluster centroids, which results in flat homogeneous regions that contribute to the defining features of cartoons.

Element-wise Edge Masking: The binary edge map generated during the second stage is combined with the quantized color image by performing an element-wise multiplication operation. This operation allows for the superposition of structural boundaries with flat shading, thereby obtaining the final result.

Convolution, edge detection, clustering, and masking operations collectively contribute to defining a balance of abstraction and structural preservation that is scalable for real-time implementation in robotic vision systems.

V. Results and Analysis

The proposed framework was tested on different real-world images, where the images were taken under different lighting conditions, scales of objects, and complexities of backgrounds. Each image was processed by the entire framework, including Toeplitz-based convolution smoothing, bilateral refinement, edge detection by the Canny operator, and K-Means quantization. One of the tested images is shown in Fig. 1.

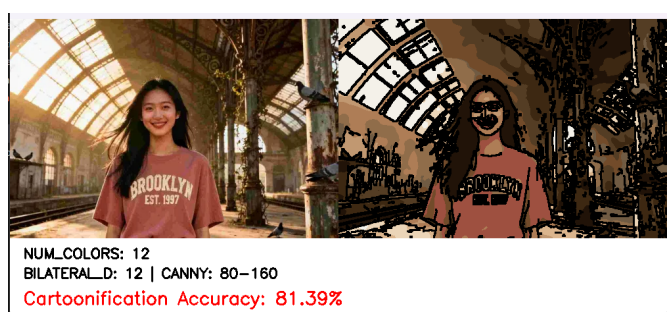

Fig. 1. Sample output of the proposed system. Left: input image. Right: cartoonized output with 12 colour clusters and edge-preserving filter (Cartoonification Accuracy: 81.39%).

A. Object Detection and Region Processing

The YOLOv8 object detection module was able to successfully locate the objects in all the test images. The weighted scoring function was able to successfully identify the most prominent region of interest in each image based on object size, proximity to the center of the frame, and confidence of object detection. Limiting the stylization process to the identified region helped in reducing the computational overhead without compromising its relevance.

B. Stylization Output

The four-stage stylization pipeline yielded visually appealing results for cartoon images. Toeplitz-based convolution effectively removed high-frequency noise without compromising spatial coherence. Bilateral filtering helped refine boundaries without introducing blur. Canny edge detection effectively enhanced prominent edges, which were then overlaid onto quantized color information to produce clear, well-defined cartoon images. Quantization of color information significantly reduced color complexity, giving rise to flat shading typical of cartoon images.

C. Quantitative Evaluation

Three numerical measures were defined to measure the quality of the stylization in an objective way.

1) Colour Simplification Ratio (CSR): Compares the number of unique colors in the pre- and post-quantization images:

$$\text{CSR} = \left(1 - \frac{N_{\text{cartoon}}}{N_{\text{original}}}\right) \times 100 \tag{10}$$

2) Edge Preservation Ratio: Measures the proportion of the original edges that remain in the resulting stylized image:

$$E_r = \frac{\text{Edges}_{\text{cartoon}}}{\text{Edges}_{\text{original}}} \times 100 \tag{11}$$

3) Structural Change (MSE-based): The structural deviation between the original and the stylized image is given by the Mean Squared Error (MSE) between the grayscale versions of the two, normalized to a bounded score S_c.

D. Composite Stylization Score

The weighted composite score combines all three metrics :

$$S = 0.4(\text{CSR}) + 0.3(E_r) + 0.3(S_c) \tag{12}$$

The stylization score for the image in Fig. 1, for example, was found to be **81.39**, reflecting a good level of abstraction and structural integrity.

E. Processing Performance

With the optimization of numerical libraries for the implementation of the Toeplitz filter and K-Means clustering, the framework can process images of standard resolution in near-real time. The architecture of the framework, based on a modular pipeline design, suggests that it could be accelerated further through the use of a GPU. The experimental validation of the framework demonstrates its effectiveness in providing a useful balance of factors.

VI. Future Work

In addition, planned extensions will allow for deployment and benchmarking of the system on resource-constrained embedded devices like NVIDIA's Jetson series. Adaptive kernel selection depending on the scene is also expected to further improve the quality of stylization under environments with highly variable lighting or texture densities. Accelerating the convolution and clustering operations with GPU support is seen as an avenue for further reducing response time. Future directions of development will look at extending the system to support multiple objects and exploring neural stylization models that are light enough for real-time requirements.

VII. Conclusion

In this paper, an autonomous image cartoonification framework is proposed that is grounded in well-defined mathematical principles for each of its stages of processing. By employing Toeplitz matrices to represent convolution operations, its linearity and shift-invariance are made explicit, offering an analytically tractable approach to smoothing that is otherwise typically performed empirically. Moreover, by employing the YOLO object detection algorithm with its proposed composite weighted scoring mechanism, intelligent region prioritization is possible that is analogous to how visual attention is implemented in autonomous robotic perception systems.

In employing the proposed four-stage stylization process comprising Toeplitz convolution filtering, gradient edge extraction, K-Means quantization, and element-wise edge masking, visually consistent cartoonified images are produced that maintain structural integrity. Quantitative validation of the proposed autonomous image cartoonification framework is performed by employing the proposed color simplification ratio, edge preservation ratio, and structural change score defined using MSE, demonstrating its effectiveness in visual abstraction without causing undue distortions to the original image. It is shown that by employing a rigorous mathematics-driven approach to image stylization, an efficient and scalable solution is possible that is well-suited for autonomous robotic perception systems.

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